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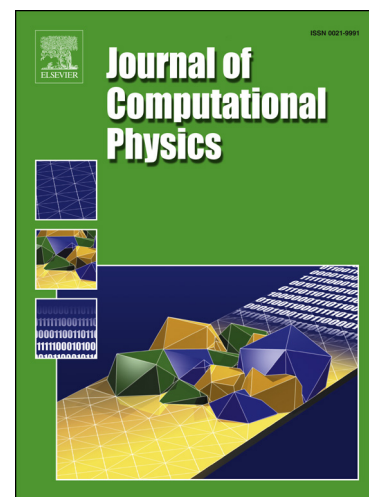
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A 2D finite volume cell-centered multi-dimensional direct ALE solver for hydrodynamics

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Abstract

In this paper we describe a second order multi-dimensional scheme, belonging to the class of direct Arbitrary Lagrangian Eulerian (ALE) methods, for the solution of non linear hyperbolic systems of conservation law. The scheme is constructed upon a cell-centered explicit Lagrangian solver completed with an edge-based upwinded formulation of the numerical fluxes, computed from the MUSCL-Hancock method, to obtain a full ALE formulation. Numerical fluxes depend on nodal grid velocities which is either set or computed to avoid most of the mesh problems typically encountered in purely Lagrangian simulations. In order to assess the robustness of the scheme, most results proposed in this paper have been obtained by computing the grid velocities as a fraction of the Lagrangian nodal velocities, the ratio being set before running the test case. The last part of the paper describes preliminary results about the triple point test case run in the ALE framework by computing the grid velocities with the fully adaptive Large Eddy Limitation (L.E.L.) method proposed in [1]. Such a method automatically computes the grid velocities at each node defining the mesh from the local characteristics of the flow. We

eventually discuss the advantages and the drawback of the coupling.

Keywords: Arbitrary Lagrangian Eulerian, ALE, Direct ALE, hydrodynamics, numerical scheme, single material, CFD, numerical simulation, fluid mechanics, GLACE, EUCCLHYD, Large Eddy Limitation

1. Introduction

Numerical simulations of fluid flow have been extensively performed on a large range of hydrodynamic phenomena over the last 60 years. The governing equations can be implemented using the finite volume approach in a given coordinate system. Several schemes have been developed and these can be divided into three categories: the first is based on a Lagrangian description of the matter for which the computational mesh follows the flow [2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13]. Mass conservation is naturally ensured in this approach, which makes Lagrangian schemes the best candidate for preserving and following interfaces in multi-material simulations. However, when the flow is subject to sharp local velocity gradients or local vorticity, mesh distortion can occur and a remeshing/remapping step is inevitable. Unfortunately, this procedure is usually expensive and the remap can be inaccurate and introduce excessive diffusion. The second category adopts an Eulerian point of view for which the computational mesh is fixed, leading to the appearance of convective terms in the governing equations, even in the mass conservation equation [14, 15, 16, 17, 18, 19]. Difficulties arise in multi-material simulations when following material interfaces. Capturing or reconstructing the interfaces can be very complex, time consuming and inaccurate (for example, the order of Young's VOF method is less than one [20]),

although significant improvements have been made recently [21, 22, 23]. In the 1970s, the combination of the best properties of the Lagrangian and Eulerian methods led to the creation of a third category, known as arbitrary Lagrangian Eulerian (ALE) methods [24]. The main feature of ALE is to decorrelate the flow motion and the mesh motion to avoid any mesh tangling which may appear in purely Lagrangian Simulations. ALE methods may be divided into two classes, *indirect* [25, 26, 27, 28, 29, 30] and *direct* [31, 32, 33, 34]. Indirect methods are implemented in three steps: (1) a Lagrangian stage in which the solution and the computational mesh are updated; (2) a rezoning stage that regularizes the tangled mesh; (3) a remapping stage in which the Lagrangian solution is transferred to the rezoned mesh. For indirect ALE the decorrelation between mesh and flow is only performed occasionally when the Lagrangian stage becomes critical and might result in simulation failure. In this paper, only direct ALE methods are considered. In these methods, the hydrodynamic equations are solved in an arbitrarily moving coordinate frame, introducing advective terms in the governing equations that depend on the grid velocity. The result is a continuous rezoning capability, with no extra time cost. Overall, ALE methods can manage greater distortions than purely Lagrangian methods but with a better accuracy than purely Eulerian approaches.

In this paper we describe a second order multi-dimensional scheme, belonging to the class of direct Arbitrary Lagrangian Eulerian (ALE) methods, named DALEC³ since it is a **D**irect **A**LE **C**ell-**C**entered **C**ode. It is used to solve the well-known Euler equations with the Geometrical Conservation

Law for the simulation of compressible hydrodynamic problems. Recall the integral form of these equations in a frame moving at the velocity $\vec{w}$ (see e.g. [34]):

$$\frac{d}{dt} \int_{V(t)} dV - \int_{S(t)} \vec{w} \cdot \vec{\nu} dS = 0, \quad (1)$$

$$\frac{d}{dt} \int_{V(t)} \rho dV + \int_{S(t)} \rho \vec{\mu} \cdot \vec{\nu} dS = 0, \quad (2)$$

$$\frac{d}{dt} \int_{V(t)} \rho \vec{v} dV + \int_{S(t)} (\rho \vec{v} \vec{\mu} \cdot \vec{\nu} + p \vec{\nu}) dS = 0, \quad (3)$$

$$\frac{d}{dt} \int_{V(t)} \rho E dV + \int_{S(t)} (\rho E \vec{\mu} \cdot \vec{\nu} + p \vec{v} \cdot \vec{\nu}) dS = 0, \quad (4)$$

$$\frac{d}{dt} (\vec{x}) = \vec{w}, \quad (5)$$

where the volume integral is computed over an arbitrary volume $V(t)$ moving at velocity $\vec{w}$. The boundary of $V(t)$ is denoted by $S(t)$, and $\vec{\nu}(t)$ denotes the unitary normal on $S(t)$ pointing outwards. Properties characterizing the fluid are the density ρ , velocity $\vec{v}$, total energy E and pressure p . The relative velocity will be denoted $\vec{\mu} = \vec{v} - \vec{w}$. System (1)-(5) is closed with an equation of state in order to compute the pressure.

The first part of the paper describes the discrete formulation of the system (1)-(5). It is constructed upon a cell-centered Lagrangian solver completed with edge-based upwinded advective fluxes to obtain a full ALE formulation. We will focus on the two cell-centered schemes GLACE and EUCCLHYD which have demonstrated their properties in references [6, 8, 9, 10]. The treatment of the boundary conditions is presented, as well as the CFL condition used to compute the time step and the second-order extension of the scheme. We also derive the entropy balance and show that our scheme satisfies the 2nd law of thermodynamics.

The second part of the paper presents several numerical results that assess the robustness and the accuracy of the scheme. To do so, most of the results have been obtained by computing the grid velocities as a fraction of the nodal Lagrangian velocities. However, our strategy is to couple the scheme with a method that adaptively computes the grid velocities. Such strategy is a common purpose for the development of direct ALE schemes, and a well-known method (coming from the indirect ALE methods community) is to rely on a rezoning procedure that keeps an acceptable geometrical mesh quality [32, 35, 36]. Recently, a promising technique, the Large Eddy Limitation method, has been developed by Costes and Ghidaglia [1]. The main feature of this technique is to express, from the Hodge decomposition, the nodal Lagrangian velocities as the sum of an irrotational part and a divergence-free part, and to limit each component regarding the local characteristic of the flow to obtain the grid velocities. Our purpose is to couple such method with the DALEC³ scheme. The last part of the paper then proposes a brief theoretical description of the L.E.L method and preliminary results obtained for the single material triple point problem. Conclusions and perspectives end the paper.

2. Description of the scheme

2.1. Notation

The computational mesh is assumed to be composed of polygons. We denote the set of nodes, cells and edges defining the mesh respectively $\mathcal{N}$, $\mathcal{C}$ and $\mathcal{E}$. In the rest of the paper, specific indices $\{j, r, k\}$ will be used

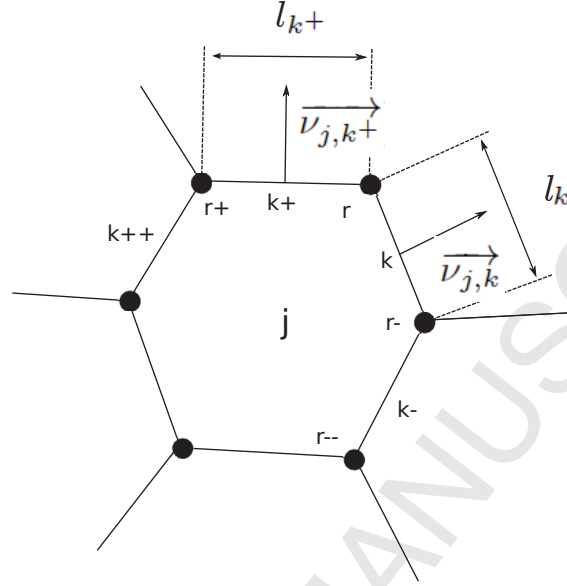


Figure 1: polygonal cell belonging to the computational mesh.

respectively to denote cells, nodes and edges.

In figure (1), we represent a 2D polygonal cell 'j' whose area will be denoted V_j . The cell is defined by the set of nodes $\mathcal{N}(j)$ and the set of edges $\mathcal{E}(j)$. The length of the edge 'k' is denoted l_k , while $\vec{\nu}_{j,k}$ is the discrete outward pointing normal to this edge. Corner vector $l_{j,r} \vec{\nu}_{j,r}$, illustrated in figure 2, are defined at each node 'r' in the cell 'j' as:

$$l_{j,r} \vec{\nu}_{j,r} = \frac{1}{2} (l_k \vec{\nu}_{j,k} + l_{k+} \vec{\nu}_{j,k+}) , \quad (6)$$

The corner vectors satisfy the two following fundamental properties. First, the sum of all corners vectors performed over the set of cells $\mathcal{C}(r)$ sharing any

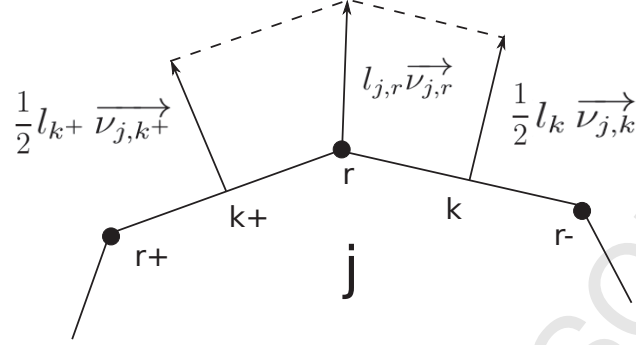


Figure 2: Definition of the corner vector $l_{j,r} \vec{\nu}_{j,r}$.

internal node ¹ is zero:

$$\sum_{j \in \mathcal{C}(r)} l_{j,r} \vec{\nu}_{j,r} = \vec{0}, \quad (7)$$

Second, the sum of all corners vectors performed over the set of nodes $\mathcal{N}(j)$ defining the cell 'j' is zero:

$$\sum_{r \in \mathcal{N}(j)} l_{j,r} \vec{\nu}_{j,r} = \vec{0}. \quad (8)$$

2.2. Lagrangian starting point

Our strategy is to start from a robust Lagrangian scheme and then add advective fluxes to obtain a full ALE formulation. The choice of the Lagrangian scheme is arbitrary providing that the compatibility between the motion of the nodes, given by (5), and the Geometrical Conservation Law (GCL) (1), is ensured. It is still unclear why this property should be satisfied, but it has been observed that numerical results were much less accurate if this compatibility was missing. As a result, we focus on GLACE and EU-CCLHYD. The discrete Lagrangian formulation of the system (1)-(5) is the

¹meaning which is not on a boundary

following:

$$\frac{d}{dt}(V_j) - \sum_{k \in \mathcal{E}(j)} \vec{v}_k \cdot l_k \vec{v}_{j,k} = 0, \quad (9)$$

$$\frac{d}{dt}(\rho_j V_j) = 0, \quad (10)$$

$$\frac{d}{dt}(\rho_j V_j \vec{v}_j) + \sum_{k \in \mathcal{E}(j)} \vec{F}_{j,k} = 0, \quad (11)$$

$$\frac{d}{dt}(\rho_j V_j E_j) + \sum_{k \in \mathcal{E}(j)} W_{j,k} = 0, \quad (12)$$

$$\frac{d}{dt}(\vec{x}_r) = \vec{v}_r, \quad (13)$$

where V_j , ρ_j , $\vec{v}_j$, E_j are respectively the volume, the density, the flow velocity and the total energy of the fluid in the cell 'j', $\vec{v}_r$ and $\vec{v}_k$ respectively denote the Lagrangian nodal velocity and the edge velocity, $\vec{F}_{j,k}$ is the pressure force exerted by the cell 'j' on the edge 'k' and $W_{j,k}$ is the pressure work. Sums are performed over the set of edges $\mathcal{E}(j)$ bounding the cell 'j'. In order to get a scheme which is second order in space and time, numerical fluxes as well as nodal velocities are computed at time $t^{n+1/2} = t^n + \frac{\Delta t}{2}$ by using the MUSCL-Hancock method, which is described in the section (2.6). The time derivative operator $\frac{d}{dt}()$ is defined as:

$$\frac{d}{dt}(\chi_j) = \frac{\chi_j^{n+1} - \chi_j^n}{\Delta t}.$$

The compatibility between the motion of the nodes and the DGCL is ensured by defining the edge velocity as the half sum of the velocities of the nodes bounding the edge [10]:

$$\forall k \in \mathcal{E}, \quad \vec{v}_k = \frac{1}{2}(\vec{v}_r + \vec{v}_{r^-}), \quad (14)$$

the edge 'k' being bounded by the nodes $\{r, r^-\}$.

2.3. Expression of the edge pressure force and the edge pressure work.

In GLACE and EUCCLHYD, the pressure force is defined at the nodes, such as:

$$\forall j \in \mathcal{C}, \forall r \in \mathcal{N}(j), \vec{F}_{j,r} = p_{j \rightarrow r} \cdot l_{j,r} \vec{\nu}_{j,r} - \overline{\overline{\sigma}}_{j,r} (\vec{v}_r - \vec{v}_{j \rightarrow r}). \quad (15)$$

where the corner vector $l_{j,r} \vec{\nu}_{j,r}$ is given by the relation (6). Quantities carrying the notation $j \rightarrow r$ are reconstructed from the cell-centered value at the node 'r' by using the MUSCL-Hancock method, see section 2.6 for details. The nodal acoustic tensor $\overline{\overline{\sigma}}_{j,r} \in \mathbb{R}^2 \times \mathbb{R}^2$ is given for both schemes in table 1. Notation $\otimes$ stands for the tensor product and $z_{r,j}$, $z_{k,j}$ and $z_{k,j}$ are the

Scheme	$\overline{\overline{\sigma}}_{j,r}$
GLACE	$z_{r,j} (l_{j,r} \vec{\nu}_{j,r} \otimes \vec{\nu}_{j,r})$
EUCCLHYD	$(z_{k,j} l_k \vec{\nu}_{j,k} \otimes \vec{\nu}_{j,k} + z_{k^+,j} l_{k^+} \vec{\nu}_{j,k^+} \otimes \vec{\nu}_{j,k^+})$

Table 1: Expression of the acoustic tensor for GLACE and EUCCLHYD schemes

generalized non-linear corner impedance designed to improve the robustness

of the scheme [10, 45]:

$$\begin{aligned} z_{r,j} &= \rho_j \left[c_j + \frac{\gamma+1}{2} |\vec{v}_r - \vec{v}_{j \rightarrow r}, \vec{\nu}_{j,r}| \right], \\ z_{k,j} &= \rho_j \left[c_j + \frac{\gamma+1}{2} |\vec{v}_r - \vec{v}_{j \rightarrow r}, \vec{\nu}_{j,k}| \right], \\ z_{k^+,j} &= \rho_j \left[c_j + \frac{\gamma+1}{2} |\vec{v}_r - \vec{v}_{j \rightarrow r}, \vec{\nu}_{j,k^+}| \right], \end{aligned}$$

with c_j the speed of sound.

For later purposes, we need to define the nodal pressure as:

$$p_{j,r} = \frac{||\vec{F}_{j,r}||}{l_{j,r}}. \quad (16)$$

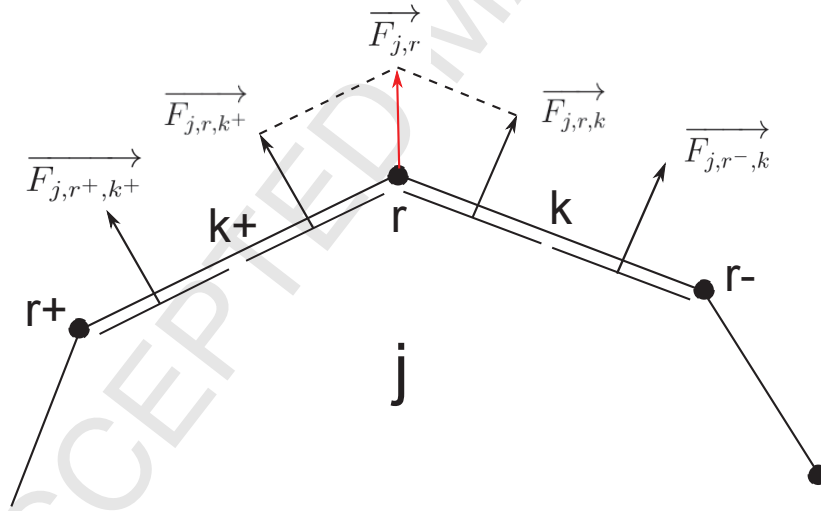


Figure 3: Representation of the nodal force on the node 'r', and of the edge forces applied on the edges connected to the node 'r'

The nodal force defined at the node 'r' may be expressed as the sum of two components applied on the surrounding edges (see fig. 3):

$$\vec{F}_{j,r} = \vec{F}_{j,r,k} + \vec{F}_{j,r,k^+}.$$

Similarly, the nodal force applying on the node 'r-1' is given by the sum:

$$\overrightarrow{F_{j,r-}} = \overrightarrow{F_{j,r-,k}} + \overrightarrow{F_{j,r-,k-}} .$$

The force that applies on the edge 'k' is then naturally equal to the sum:

$$\overrightarrow{F_{j,k}} = \overrightarrow{F_{j,r,k}} + \overrightarrow{F_{j,r-,k}} . \quad (17)$$

GLACE's first formulation [6] was constructed by assuming that the nodal pressure $p_{j,r}$ is equally applied on each half edge connected to the node 'r'.

Thus:

$$\overrightarrow{F_{j,r,k}} = p_{j,r} \cdot \frac{1}{2} l_k \overrightarrow{\nu_{j,k}} , \quad (18)$$

$$\overrightarrow{F_{j,r-,k}} = p_{j,r-} \cdot \frac{1}{2} l_k \overrightarrow{\nu_{j,k}} , \quad (19)$$

which finally gives:

$$\overrightarrow{F_{j,k}} = l_k \overrightarrow{\nu_{j,k}} \cdot \frac{1}{2} (p_{j,r} + p_{j,r-}) . \quad (20)$$

In EUCLHYD, the expression of the edge forces (18)-(19) do not use the nodal pressures but are explicitly defined:

$$\overrightarrow{F_{j,r,k}} = p_{k,r} \cdot \frac{1}{2} l_k \overrightarrow{\nu_{j,k}} , \quad (21)$$

$$\overrightarrow{F_{j,r-,k}} = p_{k,r-} \cdot \frac{1}{2} l_k \overrightarrow{\nu_{j,k}} , \quad (22)$$

with:

$$p_{k,r} = p_{j \rightarrow r} - \rho_j c_j (\overrightarrow{v_r} - \overrightarrow{v_{j \rightarrow r}} , \overrightarrow{\nu_{j,k}}) , \quad (23)$$

$$p_{k,r-} = p_{j \rightarrow r-} - \rho_j c_j (\overrightarrow{v_{r-}} - \overrightarrow{v_{j \rightarrow r-}} , \overrightarrow{\nu_{j,k}}) . \quad (24)$$

Finally, the explicit expression of the pressure force in the EUCLHYD scheme is:

$$\overrightarrow{F_{j,k}} = l_k \overrightarrow{\nu_{j,k}} \cdot \frac{1}{2} (p_{k,r} + p_{k,r-}) . \quad (25)$$

Remark: in (23)-(24), the values of the pressure $p_{j \rightarrow r}$ and the velocity $\overrightarrow{v_{j \rightarrow r}}$ are reconstructed at the nodes, even if the pressures $p_{k,r}$ and $p_{k,r-}$ apply on the edge 'k' of the cell, in order to preserve the conservation of momentum and total energy in the second-order extension of the scheme.

We end this section by giving the expression of the pressure work in both schemes:

$$W_{j,k} = \overrightarrow{F_{j,r,k}} \cdot \overrightarrow{v_r} + \overrightarrow{F_{j,r-,k}} \cdot \overrightarrow{v_{r-}}, \quad (26)$$

where, explicitly:

$$\begin{aligned} W_{j,k} &= l_k \overrightarrow{\nu_{j,k}} \cdot \frac{1}{2} (p_{j,r} \overrightarrow{v_r} + p_{j,r-} \overrightarrow{v_{r-}}), \text{ in GLACE,} \\ W_{j,k} &= l_k \overrightarrow{\nu_{j,k}} \cdot \frac{1}{2} (p_{k,r} \overrightarrow{v_r} + p_{k,r-} \overrightarrow{v_{r-}}), \text{ in EUCCLHYD.} \end{aligned}$$

2.4. Nodal Solver

A nodal solver is used to compute the nodal velocities $\overrightarrow{v_r}$, $\forall r \in \mathcal{N}$ from the cell-centered velocities [9, 10, 37]. It expresses that the sum of all forces around any internal node 'r' is zero:

$$\sum_{j \in \mathcal{C}(r)} \overrightarrow{F_{j,r}} = \overrightarrow{0}. \quad (27)$$

From the relation (15), it turns into the 2×2 non-linear system:

$$\sum_{j \in \mathcal{C}(r)} [p_{j \rightarrow r} \cdot l_{j,r} \overrightarrow{\nu_{j,r}} - \overline{\overline{\sigma_{j,r}}} (\overrightarrow{v_r} - \overrightarrow{v_{j \rightarrow r}})] = 0. \quad (28)$$

which is solved by using a newton procedure. The solution is unique since $\overline{\overline{\sigma_{j,r}}}$ is a symmetric positive-definite matrix. For the boundary nodes, the

previous solver is still valid provided some minor modifications depending on the applied boundary condition. Readers can refer to section 2.7 to read more about the implementation of the boundary conditions.

2.5. ALE formulation of the scheme

The full ALE formulation is obtained by adding upwinded advective fluxes to the Lagrangian scheme (9)-(12)

$$\frac{d}{dt} (V_j) - \sum_{k \in \mathcal{E}(j)} \vec{w}_k \cdot l_k \vec{\nu}_{j,k} = 0, \quad (29)$$

$$\frac{d}{dt} (\rho_j V_j) + \sum_{k \in \mathcal{E}(j)} \rho_k^{\text{up}} \vec{\mu}_k \cdot l_k \vec{\nu}_{j,k} = 0, \quad (30)$$

$$\frac{d}{dt} (\rho_j V_j \vec{v}_j) + \sum_{k \in \mathcal{E}(j)} \left(\rho_k^{\text{up}} \vec{v}_k^{\text{up}} \vec{\mu}_k \cdot l_k \vec{\nu}_{j,k} + \vec{F}_{j,k} \right) = 0, \quad (31)$$

$$\frac{d}{dt} (\rho_j V_j E_j) + \sum_{k \in \mathcal{E}(j)} (\rho_k^{\text{up}} E_k^{\text{up}} \vec{\mu}_k \cdot l_k \vec{\nu}_{j,k} + W_{j,k}) = 0, \quad (32)$$

$$\frac{d \vec{x}_r}{dt} = \vec{w}_r. \quad (33)$$

where ρ_k^{up} , $\vec{v}_k^{\text{up}}$ and E_k^{up} stand for the upwind density, velocity and energy, which are chosen regarding the sign of the quantities $\vec{\mu}_k \cdot \vec{\nu}_{j,k}$. Recall $\chi_k^{\text{up}} = \{\rho_k^{\text{up}}, \vec{v}_k^{\text{up}}, E_k^{\text{up}}\}$, we get, in the case of an edge 'k' shared by two cells $\{j, j+1\}$ (for a boundary edge, see section 2.7):

$$\chi_k^{\text{up}} = \frac{1}{2} (\chi_{j \rightarrow k} + \chi_{j+1 \rightarrow k}) + \frac{1}{2} \text{sign}(\vec{\mu}_k \cdot \vec{\nu}_{j,k}) (\chi_{j \rightarrow k} - \chi_{j+1 \rightarrow k}), \quad (34)$$

$\text{sign}(x) \in \{-1, 0, 1\}$ is the sign of the real number x . Terms $\chi_{j \rightarrow k}$ and $\chi_{j+1 \rightarrow k}$ respectively denote values computed at the edge from a piecewise linear reconstruction within the cells 'j' and 'j+1', see (2.6) for details.

The compatibility between the motion of the nodes (33) and the Geometrical Conservation Law (29) is ensured by computing the edge grid velocities as the half sum of the nodes velocities:

$$\forall k \in \mathcal{E}, \quad \vec{w}_k = \frac{1}{2} (\vec{w}_r + \vec{w}_{r^-}) . \quad (35)$$

The edge relative velocity $\vec{\mu}_k$ is defined as:

$$\vec{\mu}_k = \vec{v}_k - \vec{w}_k = \frac{1}{2} (\vec{\mu}_r - \vec{\mu}_{r^-}) . \quad (36)$$

2.6. Second-order extension of the scheme

The second order extension of the scheme is obtained by using the MUSCL-Hancock method [18]. We need to compute the terms $p_{j \rightarrow r}$ and $\vec{v}_{j \rightarrow r}$ involved in the expression of the edge Force (20) or (25) and pressure work (26), as well as the upwinded quantities $\{\rho^{\text{up}}, \vec{v}^{\text{up}}, E^{\text{up}}\}$. In the following, we describe the implementation of the method for a specific cell 'j' within the mesh:

1. density, pressure, velocity and energy gradients are computed from the least-square method:

$$\left[\vec{\nabla} \chi \right]_j = \arg \min_{(\vec{\nabla} \chi) \in \mathbb{R}^2} \sum_{l \in \mathcal{C}(j)} \left[\chi_l - \left(\chi_j + \vec{\nabla} \chi \cdot (\vec{x}_l - \vec{x}_j) \right) \right]^2 , \quad (37)$$

where $\chi = \{\rho, \vec{v}, p, E\}$. The sum is performed over the set of cells neighboring the cell 'j' in which gradients are computed.

2. Gradients are used to compute boundary extrapolated quantities (at the edges and the nodes of the cell 'j') from the piecewise linear reconstruction:

$$\begin{aligned} \chi_{j \rightarrow r} &= \chi_j^n + L(\chi)_j \left[\vec{\nabla} \chi \right]_j^n \cdot (\vec{x}_r - \vec{x}_j^n) , \\ \chi_{j \rightarrow k} &= \chi_j^n + L(\chi)_j \left[\vec{\nabla} \chi \right]_j^n \cdot (\vec{x}_k - \vec{x}_j^n) . \end{aligned}$$

The edge midpoint position is computed as $\vec{x}_k = \frac{1}{2} (\vec{x}_r + \vec{x}_{r^-})$, $\vec{x}_j^n$ is the cell center at time t^n computed from the relation:

$$\vec{x}_j^n = \frac{1}{V_j} \int_{V_j} \vec{x}(t^n) dV,$$

and $L(\chi)_j$ are slope limiters computed from the Barth-Jespersen procedure [39], using the limiter proposed by Venkatakrishnan [10]. Although the choice of the limiter is arbitrary, the one proposed by Venkatakrishnan gives excellent results in terms of stabilization in all the test cases proposed in section 3.

3. Upwind values ρ_k^{up} , $\vec{v}_k^{\text{up}}$, E_k^{up} are computed from (34).
4. Nodal extrapolated values $\chi_{j \rightarrow r}$ and upwind values are evolved by the half time step $\frac{\Delta t}{2}$ by substituting it for the cell-centered values in (29)-(32) and using values computed in steps 2 and 3 in the expression of the numerical fluxes. Then, the edge force and pressure work at time $t^{n+1/2}$ are computed.
5. Edge force, pressure work and upwind values computed at time $t^{n+1/2}$ are used in the system (29)-(33) to compute the numerical solution at time t^{n+1} .

2.7. Boundary conditions

On the mesh boundary $\partial\mathcal{M}$, we must provide the values of the upwind parameters ρ_k^{up} , $\vec{v}_k^{\text{up}}$, E_k^{up} and compute the value of the nodal velocity $\vec{v}_r$. In this section, we present the conditions that have been used in the results proposed in section 3.

Sliding on a plane wall

Let $\overrightarrow{\nu_{\text{wall}}}$ be the outward pointing normal to the wall. The sliding condition for any sliding node is $\overrightarrow{v_r} \cdot \overrightarrow{\nu_{\text{wall}}} = 0$. In this case, the nodal balance (27) must take into account the reaction force of the wall $\overrightarrow{R_{\text{wall}}} = |R_{\text{wall}}| \overrightarrow{\nu_{\text{wall}}}$:

$$\sum_{j \in \mathcal{C}(r)} \overrightarrow{F_{j,r}} + \overrightarrow{R_{\text{wall}}} = 0. \quad (38)$$

We eliminate the unknown $\overrightarrow{R_{\text{wall}}}$ by taking the scalar product of (38) and $\overrightarrow{\nu_{\text{wall}}}^\perp$, which leads to the system:

$$\begin{cases} \overrightarrow{v_r} \cdot \overrightarrow{\nu_{\text{wall}}} = 0, \\ \sum_{j \in \mathcal{C}(r)} [p_{j \rightarrow r} \cdot l_{j,r} \overrightarrow{\nu_{j,r}} \cdot \overrightarrow{\nu_{\text{wall}}} - \overrightarrow{\sigma_{j,r}^*} \cdot (\overrightarrow{v_r} - \overrightarrow{v_{j \rightarrow r}})] = 0, \end{cases} \quad (39)$$

with $\overrightarrow{\sigma_{j,r}^*} = (\overrightarrow{\sigma_{j,r}})^t \overrightarrow{\nu_{\text{wall}}}^\perp$. Exponent t denotes the transpose. Solving this system gives the velocity $\overrightarrow{v_r}$ for each sliding node.

The choice of the upwind values are unnecessary in this case, given the following proposition.

Proposition 1. *For any boundary edge carrying a wall condition, we have:*

$$\overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} = 0. \quad (40)$$

Proof. For a node 'r' sliding on the plane wall, the sliding condition imposes that $\overrightarrow{v_r} \cdot \overrightarrow{\nu_{\text{wall}}} = 0$ as well as $\overrightarrow{w_r} \cdot \overrightarrow{\nu_{\text{wall}}} = 0$. As a result, we have $\overrightarrow{\mu_r} \cdot \overrightarrow{\nu_{\text{wall}}} = 0$. Now consider two consecutive nodes $\{r^-, r^+\}$ sliding on the same plane and defining the edge 'k'. Obviously, $\overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{\text{wall}}} = 0$ from (36). Moreover, in this case, we naturally have $\overrightarrow{\nu_{\text{wall}}} = -\overrightarrow{\nu_{j,k}}$. $\square$

As a result, the advective terms in (30), (31) and (32) are zero whatever the values of the upwinded numerical fluxes.

Normal velocity condition

This condition can be seen as a sliding condition where the wall is moving at the velocity $\vec{v}_i(t)$ in the direction of the outward-pointing normal $\vec{\nu}_{\text{wall}}$. The first relation in the system (39) is substituted for the following continuity condition:

$$\vec{v}_r \cdot \vec{\nu}_{\text{wall}}^\perp = \vec{v}_i(t) \cdot \vec{\nu}_{\text{wall}}^\perp.$$

The system to solve is then:

$$\begin{cases} \vec{v}_r \cdot \vec{\nu}_{\text{wall}}^\perp = \vec{v}_i(t) \cdot \vec{\nu}_{\text{wall}}^\perp, \\ \sum_{j \in \mathcal{C}(r)} [p_{j \rightarrow r} \cdot l_{j,r} \vec{\nu}_{j,r} \cdot \vec{\nu}_{\text{wall}} - \vec{\sigma}_{j,r}^* \cdot (\vec{v}_r - \vec{v}_{j \rightarrow r})] = 0. \end{cases} \quad (41)$$

As in the previous case, the upwind choice is unnecessary.

Pressure condition

Consider a set of nodes on which an external pressure force is applied. This force is expressed as $p_{\text{ext}} \vec{C}_{\text{ext}}$, with p_{ext} the value of the external pressure and $\vec{C}_{\text{ext}}$ the direction of application of the force. Recalling eq. (27) and taking into account the external force:

$$\sum_{j \in \mathcal{C}(r)} \vec{F}_{j,r} + p_{\text{ext}} \vec{C}_{\text{ext}} = 0. \quad (42)$$

and using (7) to define the vector $\vec{C}_{\text{ext}}$:

$$\vec{C}_{\text{ext}} = - \sum_{j \in \mathcal{C}(r)} l_{j,r} \vec{\nu}_{j,r}. \quad (43)$$

the nodal solver turns into:

$$\sum_{j \in \mathcal{C}(r)} [(p_{j \rightarrow r} - p_{\text{ext}}) \cdot l_{j,r} \overrightarrow{\nu_{j,r}} - \overline{\sigma_{j,r}} (\overrightarrow{v_r} - \overrightarrow{v_{j \rightarrow r}})] = \overrightarrow{0}. \quad (44)$$

The method to solve this system and find the nodal velocity is the one proposed in the section (2.4) by substituting $p_{j \rightarrow r}$ for $p_{j \rightarrow r} - p_{\text{ext}}$.

2.8. Entropy condition

We show in this section that our first-order scheme satisfies a local entropy inequality. We define an entropy of the domain $V(t)$ as being:

$$\int_{V(t)} \rho s \, dV,$$

where s is the entropy function in the domain, and its semi-discrete finite-volume formulation in the j^{th} cell is:

$$s_j = \frac{1}{V_j} \int_{V_j} s \, dV. \quad (45)$$

Proposition 2. *The semi-discrete formulation of the entropy balance is:*

$$\begin{aligned} T_j \frac{d(\rho_j V_j s_j)}{dt} + T_j \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{up} s_k^{up} \overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} = \\ - \frac{1}{2} \sum_{k \in \mathcal{E}(j)} l_k \left(\overrightarrow{v_k^{up}} - \overrightarrow{v_j} \right)^2 \overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} + \sum_{r \in \mathcal{N}(j)} \overline{\sigma_{j,r}} (\overrightarrow{v_r} - \overrightarrow{v_j}) \cdot (\overrightarrow{v_r} - \overrightarrow{v_j}). \end{aligned} \quad (46)$$

where T_j is the temperature in the cell ' j '.

The derivation is reported in the Appendix A.

The first term on the right-hand side of (46) is either zero ($\overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} \geq 0$) or positive ($\overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} < 0$). The second term is zero or positive since the acoustic tensor is a symmetric positive-definite matrix. The following proposition is straightforward.

Proposition 3. *In any cell 'j', our scheme produces entropy in the sense that:*

$$\forall j \in \mathcal{C}(r), \quad T_j \frac{d(\rho_j V_j s_j)}{dt} + T_j \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{up} s_k^{up} \vec{\mu}_k \cdot \vec{\nu}_{j,k} \geq 0. \quad (47)$$

2.9. Stability - CFL Condition

A CFL condition is needed because the method is explicit [5, 18, 40]. In the Lagrangian Framework, we use the condition:

$$\text{CFL} = \max_{j \in \mathcal{C}} \left(\frac{c_j}{\Delta x_j} \right) \Delta t < 1, \quad (48)$$

where c_j is the speed of sound in the cell 'j', and Δx_j is an arbitrary length computed hereafter as the ratio of the volume over the smallest edge area among the surrounding edges bounding the j^{th} cell:

$$\Delta x_j = \frac{V_j}{\min_{k \in \mathcal{E}(j)} l_k}. \quad (49)$$

In the ALE Framework, we complete this formula with the the sup norm $|\cdot|_\infty$ of the flow velocity:

$$\text{CFL} = \max_{j \in \mathcal{C}} \left(\frac{|\vec{v}|_\infty + c_j}{\Delta x_j} \right) \Delta t < 1, \quad (50)$$

3. Results

We propose in this section various 2D results to show the robustness of the scheme. Our methodology of validation acts in two phases. First, the grid velocity is computed as a fraction of the Lagrangian velocity to assess the robustness of the ALE scheme. Second, the grid velocity is given by a genuine rezone strategy [1] in order to prove that our approach can be coupled to any artificial mesh motion.

3.1. Sod Problem

The well-known Sod problem is a 1D textbook case [41]. A fluid initially at rest in a shock tube is divided into two distinct regions that are separated by a virtual interface, located at the position x_i . The states on both sides of the interface are constant initially. The left state ($x \leq x_i$) is characterized by $(\rho_L, p_L) = (1, 1)$, while the right state ($x > x_i$) is defined by $(\rho_R, p_R) = (0.125, 0.1)$. The fluid is described with the ideal gas law with the specific heat ratio $\gamma = 7/5$. The computational domain is defined by $(x, y) \in [0, 1] \times [0, 0.1]$ and the mesh is initially a Cartesian grid with 150×15 square cells. The interface position is set to $x_i = 0.5$. Boundary conditions are wall conditions, except for the corner nodes whose velocity is zero at any time. For this case, we set $\text{CFL} = 0.5$.

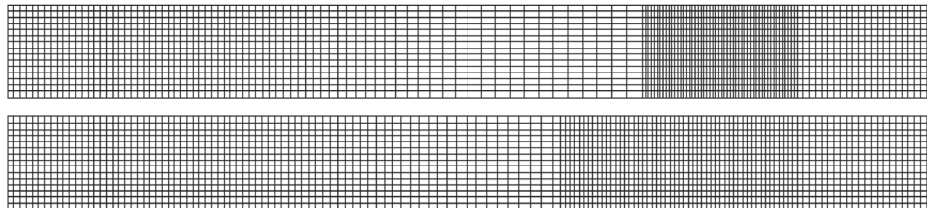


Figure 4: Sod problem. Mesh in the Lagrangian framework (top) and in ALE framework (bottom) at $t_{\text{final}} = 0.2$

The test case is run in the three systems of coordinates : Eulerian, Lagrangian and ALE. We choose here to compute the grid velocities such as $\forall r \in \mathcal{N}, \vec{w}_r = 0.5\vec{v}_r$. Computational meshes are given in the figure (4).

The numerical solutions computed at time $t = 0.2$ match the analytical solution as seen in the figures 5, 6 and 7. The contact discontinuity and the

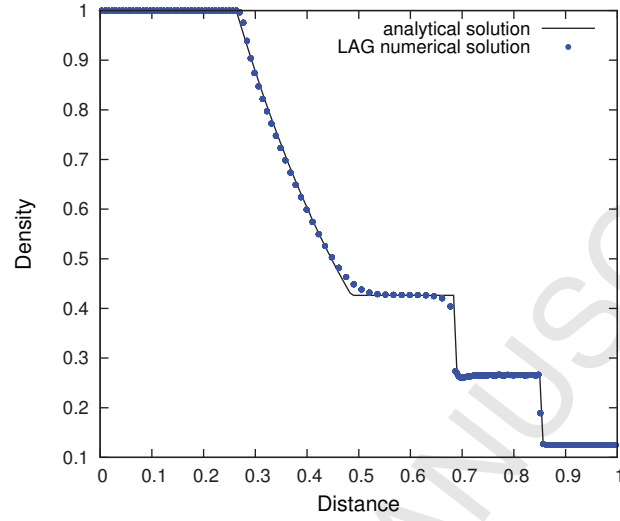


Figure 5: Comparison of the Lagrangian density profile computed at $t_{\text{final}} = 0.2$ with the analytical solution

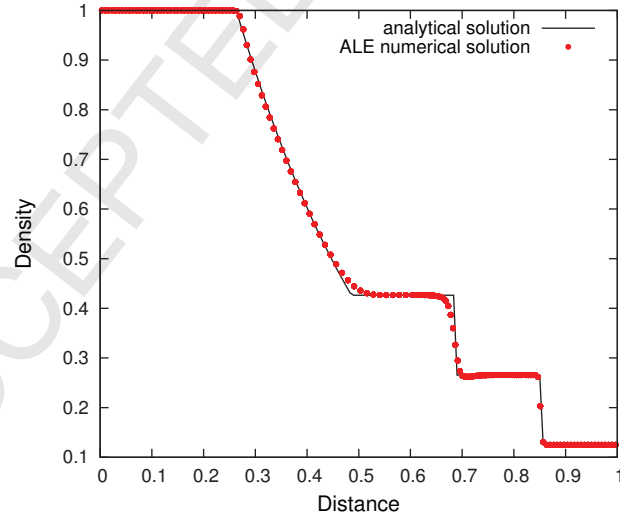


Figure 6: Comparison of the Eulerian density profile computed at $t_{\text{final}} = 0.2$ with the analytical solution

shock front are captured at the expected position, and the density plateaus amplitude match the analytical values. It is seen in the figure (8) that the accuracy of the numerical solution computed in the ALE framework lies between the accuracy of the Eulerian solution and the Lagrangian solution, by comparing the capture of the contact discontinuity. The Eulerian contact discontinuity is more diffusive, while the Lagrangian one is sharper.

3.2. Isentropic Compression - 2D Kidder problem

The 2D Kidder problem is usually run in order to assess the ability of a Lagrangian scheme to properly compute an isentropic compression. Results proposed in references [9, 10] show that both GLACE and EUCCLHYD schemes do manage to simulate this test case, and the purpose of this section is not to reach this conclusion again. Instead, we run it to compute the order the scheme in the Lagrangian Framework.

The 2D Kidder problem describes the 2D isentropic compression of a ring filled with a perfect gas [42]. From now, all quantities subscripted with the numbers 1 and 2 are attached to the inner and the outer ring. The computational domain Ω is defined in polar coordinates by $(r, \theta) \in [r_1, r_2] \times [0, \frac{\pi}{2}]$, where $r_1 = 0.9$ is the inner ring radius and $r_2 = 1$ the outer ring radius. Boundary conditions on the lateral edges ($\theta = 0$ and $\theta = \pi/2$) are sliding walls, while a pressure condition is continuously imposed on both inner and outer frontiers, with:

$$\begin{aligned} p_1(t) &= p^0(r_1) \cdot h(t)^{-\frac{2\gamma}{\gamma-1}}, \\ p_2(t) &= p^0(r_2) \cdot h(t)^{-\frac{2\gamma}{\gamma-1}}. \end{aligned}$$

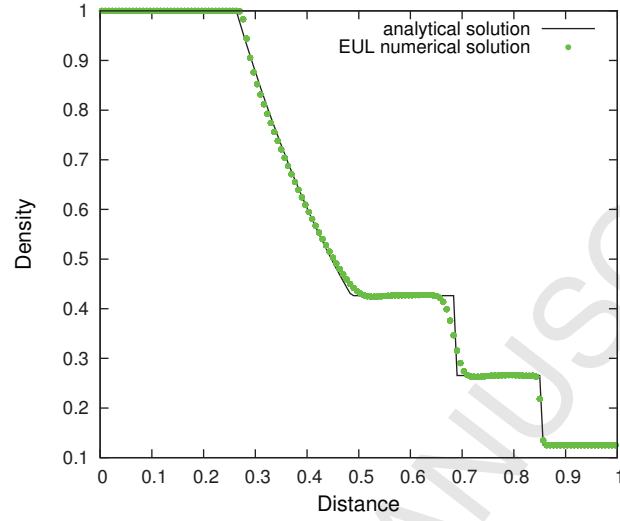


Figure 7: Comparison of the ALE density profile computed at $t_{\text{final}} = 0.2$ with the analytical solution

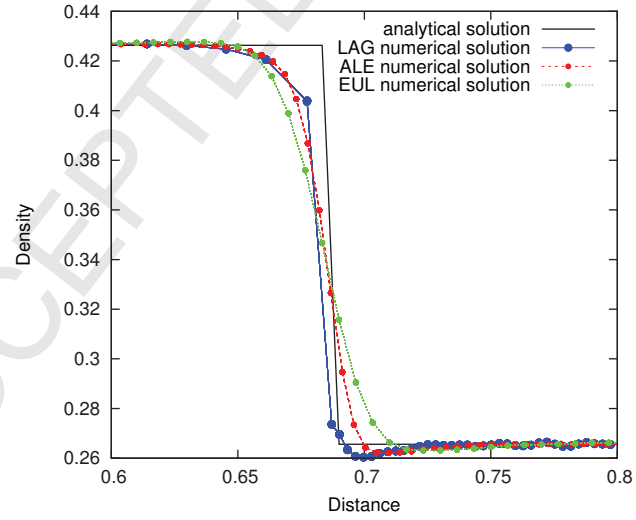


Figure 8: Comparison of the density profiles computed in the Lagrangian and ALE frameworks at $t_{\text{final}} = 0.2$

The homothety rate $h(t) = \sqrt{1 - \frac{t^2}{\tau^2}}$, $\forall t \in [0, \tau]$ depends on the focalization time τ which is defined as:

$$\tau = \sqrt{\frac{(\gamma - 1)(r_2^2 - r_1^2)}{2(c_2^2 - c_1^2)}}.$$

where c_1 and c_2 are the speeds of sound computed at the inner and outer layers.

The gas is initially at rest and its density is set to:

$$\rho^0(r) = \left(\frac{r_2^2 - r^2}{r^2 - r_1^2} \rho_1^{\gamma-1} + \frac{r^2 - r_1^2}{r_2^2 - r_1^2} \rho_2^{\gamma-1} \right)^{\frac{1}{\gamma-1}},$$

with $\rho_1 = 1$, $\rho_2 = 2$ and the adiabatic index γ is set to 2. Pressure is initialized through the adiabatic gas law:

$$p^0(r) = s (\rho^0(r))^\gamma.$$

where the isentropic coefficient is defined as $s = \frac{p_2}{\rho_2^\gamma}$. With such initial conditions, the solution is known to be an isentropic compression ($s = 1$) such that the position at time $t > 0$ is $R(t) = h(t) r$. The test is run with CFL = 0.1 until the final time $t_{\text{final}} = \frac{\sqrt{3}}{2} \tau$.

To compute the order of the scheme, we run the test case with five meshes: $\mathcal{M}_1$ is a 20×20 cells mesh, that is 20 sectors and 20 layers; $\mathcal{M}_2$ contains 40×40 cells; $\mathcal{M}_3$ contains 80×80 cells; $\mathcal{M}_4$ contains 120×120 cells and finally $\mathcal{M}_5$ contains 160×160 cells. Both initial and final meshes are given in figure (9) for a test run with the mesh $\mathcal{M}_1$. For this test, we compare the analytical position of the external and internal boundaries with the numerical ones in figure (10). We measure the rate of convergence of the numerical solution to

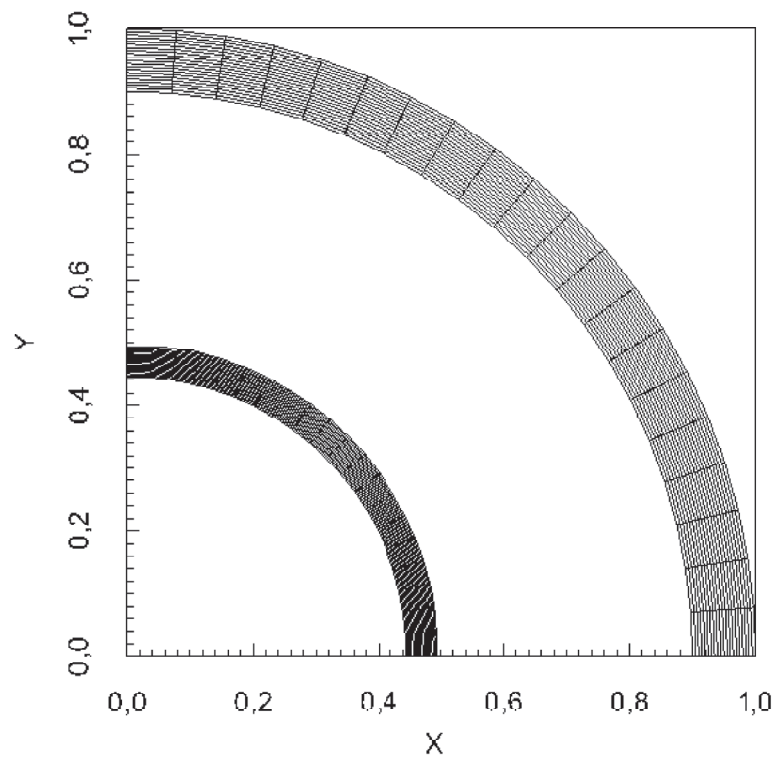


Figure 9: Computational mesh plotted at times $t = 0$ and $t = t_{\text{final}}$.

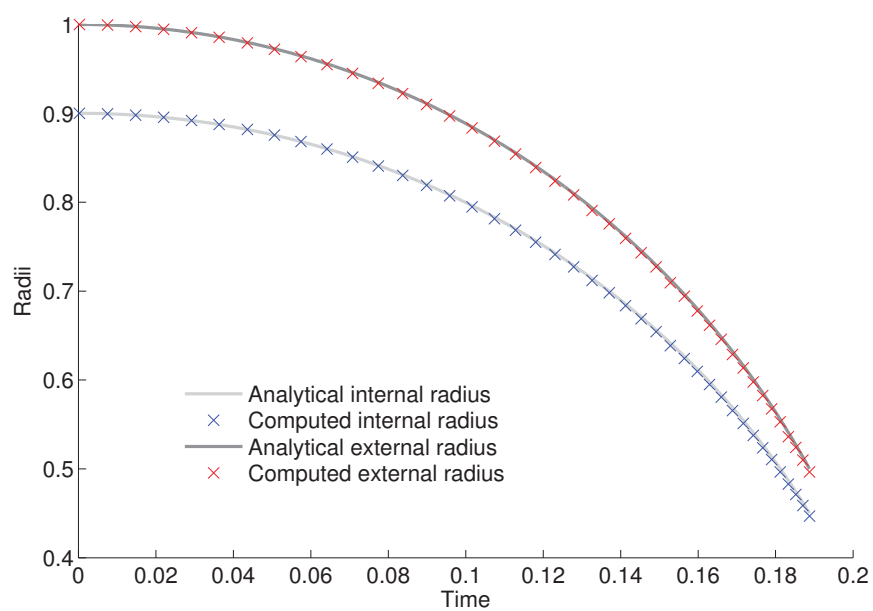


Figure 10: Comparison between analytical and radii computed with the second order extension.

the exact one in 2D by computing the L^2 norm of the difference between the exact and the computed radii as a function of the mesh resolution. Results are recorded in table 3. The experimental mean order of the convergence is ≈ 1.7 for the internal radius and ≈ 1.48 for the external radius.

Mesh	inner		outer	
	L^2	order	L^2	order
$\mathcal{M}_1$	3.7×10^{-3}	—	3.9×10^{-3}	—
$\mathcal{M}_2$	10^{-3}	1.81	1.2×10^{-3}	1.65
$\mathcal{M}_3$	3.56×10^{-4}	1.55	4.76×10^{-4}	1.37
$\mathcal{M}_4$	1.79×10^{-4}	1.69	2.72×10^{-4}	1.38
$\mathcal{M}_5$	1.09×10^{-4}	1.73	1.75×10^{-4}	1.52

Table 2: Rate of convergence of the numerical solution towards the exact solution, in function of the mesh.

3.3. 2D isentropic vortex

The 2D isentropic vortex problem [43] is run to compute the accuracy of our scheme in the Eulerian framework. A squared computational domain $\Omega = [0, 10] \times [0, 10]$ is filled with a gas characterized by the initial state $\rho_\infty = 1$, $\vec{v}_\infty = (1, 1)^t$, and a normalized ambient temperature $T_\infty^* = 1$. The latter is perturbed at the initial time by a vortex centered at $(x_{\text{vortex}}, y_{\text{vortex}}) = (5, 5)$ and characterized by $\vec{v} = \vec{v}_\infty + \vec{\delta v}$ and $T^* = T_\infty^* + \delta T^*$, where:

$$\begin{aligned} \vec{\delta v} &= \begin{pmatrix} -(y - y_{\text{vortex}}) \frac{\beta}{2\pi} \exp\left(\frac{1-r^2}{2}\right) \\ (x - x_{\text{vortex}}) \frac{\beta}{2\pi} \exp\left(\frac{1-r^2}{2}\right) \end{pmatrix}, \\ \delta T^* &= -\frac{(\gamma - 1) \beta^2}{8\gamma\pi^2} \exp(1 - r^2) . \\ r &= \sqrt{(x - x_{\text{vortex}})^2 + (y - y_{\text{vortex}})^2} \end{aligned}$$

The adiabatic index γ is set to 1.4. Parameter β , equal to 5, is used here to drive the vortex strength. The density is computed from the isentropic equation of state:

$$\rho = \rho_\infty \left(\frac{T^*}{T_\infty^*} \right)^{\frac{1}{\gamma-1}} .$$

The numerical solution computed at $t_{\text{final}} = 1$ should remain close to the smooth initial solution shifted by the ambient flow. Therefore, an effective high order of accuracy must be reached for this problem. When running this test case, periodic boundary conditions are prescribed on the domain boundaries, and CFL condition is set to 0.5.

In figure 11, we propose the density map, centered on the vortex eye, computed at the final time in both Lagrangian and Eulerian frameworks. Convergence rates are computed in the Eulerian framework by measuring L^2 errors norms between the discrete solution and the exact solution for the density at the final time while the test is run on the five following meshes: $\mathcal{M}_1 = 40 \times 40$ cells, $\mathcal{M}_2 = 80 \times 80$ cells, $\mathcal{M}_3 = 120 \times 120$ cells, $\mathcal{M}_4 = 160 \times 160$ cells, $\mathcal{M}_5 = 320 \times 320$ cells. Results are reported in Table 1, showing that the mean rate of convergence is 2.2.

Mesh	inner	
	L^2	order
$\mathcal{M}_1$	3.4×10^{-3}	—
$\mathcal{M}_2$	7.04×10^{-4}	2.13
$\mathcal{M}_3$	2.84×10^{-4}	2.18
$\mathcal{M}_4$	1.52×10^{-4}	2.24
$\mathcal{M}_5$	3.47×10^{-5}	2.26

Table 3: Rate of convergence of the numerical solution towards the exact solution, in function of the mesh.

3.4. Sedov blast wave problem on a cartesian grid

The Sedov problem models a point-blast propagating in an uniform medium with cylindrical symmetry [44]. The computational mesh is a 30×30 cartesian grid defined on the domain $[0, 1.2] \times [0, 1.2]$. A wall condition is applied at the boundaries. The fluid satisfies the ideal gas law and the specific heat ratio is equal to $7/5$. The initial velocity is $\vec{v}^0 = \vec{0}$. Initial physical properties of the medium are $(\rho^0, p^0) = (1, 10^{-6})$, except in the bottom-left cell whose initial pressure is given by the ideal gas law:

$$P_H = (\gamma - 1) \rho^0 \frac{e_H}{V},$$

where e_H is the amount of released energy and V is the volume in which the energy is released. Setting $e_H = 0.244816$ results in the propagation of a diverging shock whose front is located at radius $R = 1$ at time $t = 1$.

For the simulation of the test case, we choose EUCCLHYD as the Lagrangian scheme, with $CFL = 0.1$. The grid velocities are limited to $\forall r \in \mathcal{N}$, $\vec{w}_r = 0.7\vec{v}_r$. As in the Sod shock tube test case, we try to obtain a smoother mesh in the expansion fan.

Figure 4 presents the mesh (left) and the density map (right) computed in all frameworks at time $t_{\text{final}} = 1$. Whatever the system of coordinates, the scheme is able to preserve the symmetry of the problem and the peak location is captured at the right position. As expected, the Lagrangian solution still remains the most accurate. The points dispersion is increased for the ALE solution, except maybe in the expansion ($x \leq 0.5$) fan where the refinement

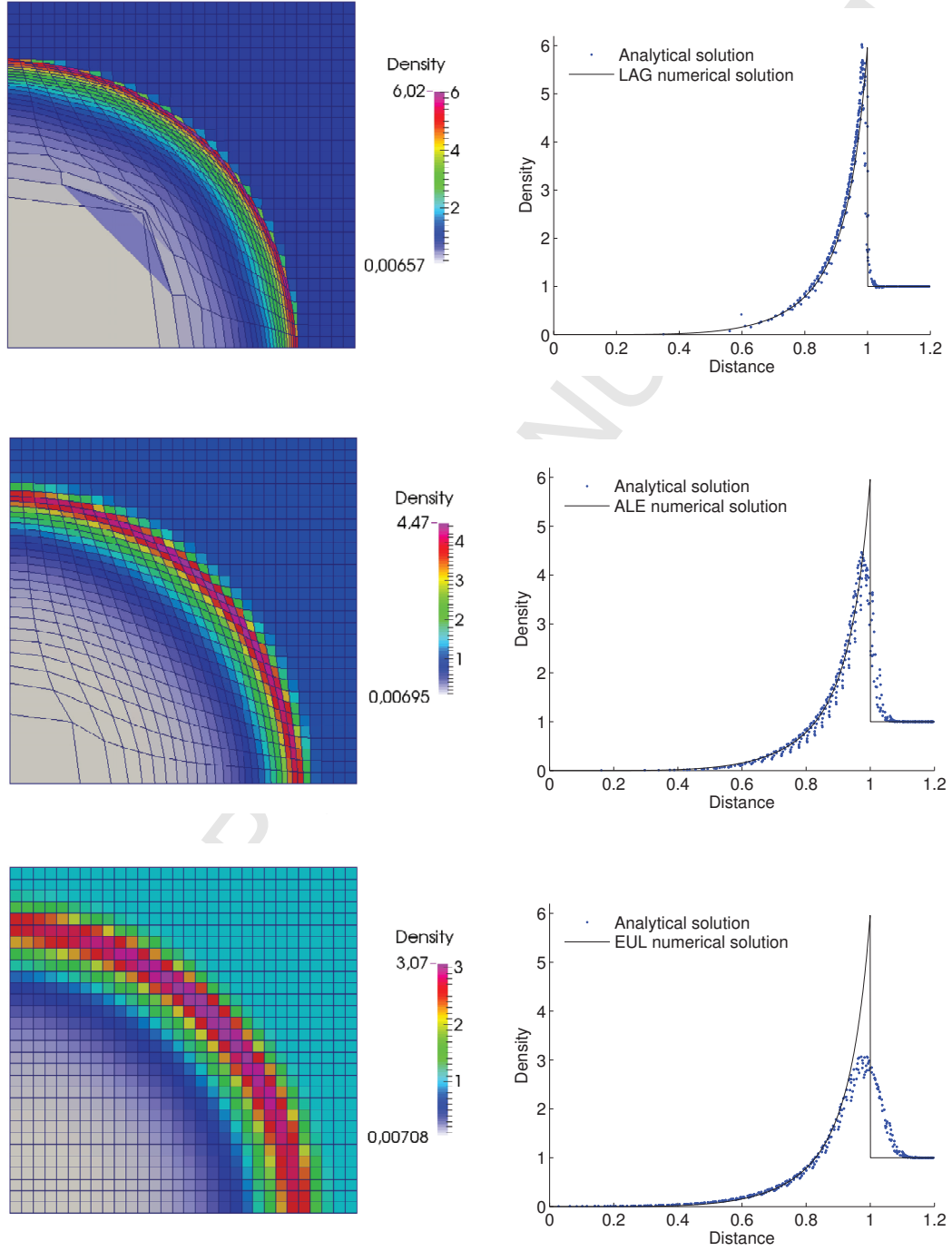


Table 4: Mesh and density map computed with EUCCLHYD scheme at the computational time $t_{\text{final}} = 1$ for the Lagrangian framework (top) and the ALE framework (bottom) in the Sedov blast wave test case

induces a fine description of the solution.

3.5. Saltzman problem

Described in [45], this problem tests the robustness of Lagrangian schemes in situations where the mesh is not representative of the flow. It consists of a strong piston-driven shock propagating in the domain $[0, 1] \times [0, 0.1]$ for which the computational mesh is initially nonuniform. The skewed initial mesh, displayed at the top of the figure 12, is obtained from a regular 100×10 Cartesian grid with the mapping:

$$\begin{cases} x_s = x + (0.1 - y) \sin(\pi x) \\ y_s = y \end{cases}.$$

Initial conditions are $(\rho, \vec{v}, e) = (1, (0, 0)^t, 10^{-6})$. On the left boundary, a normal velocity condition with $\vec{v}_i = (1, 0)^t$ and $\vec{d} = (1, 0)^t$ is prescribed, while, on the other three boundaries, a sliding condition is applied. The fluid satisfies the ideal gas law with $\gamma = 5/3$. For these initial conditions, the analytical solution to the problem is a one-dimensional infinite strength shock wave that moves at speed $D = 4/3$ to the right. The shock front hits the right wall ($x = 1$) at time $t = 0.75$. Behind the shock front, the density is equal to 4.

In the ALE framework, grid velocities are computed from the relation:

$$\vec{w}_r(t) = \begin{pmatrix} \frac{1 - x_r(t)}{1 - t} \\ 0 \end{pmatrix},$$

where t is the computational time and x_r the x-component of the nodal position $\vec{x}_r$.

The density maps obtained at the computational time $t = 0.7$ with GLACE in both Lagrangian and ALE frameworks are proposed in figures 13 and 14. In the Lagrangian framework, a spurious rotation is observed near the left boundary, which is certainly caused by a too weak diffusion in the Discrete Geometrical Conservation Law (this is not observed for instance with EUCCLHYD which is known to be much more diffusive than GLACE). In the ALE framework, we obtain a smoother mesh.

The density profiles obtained in both frameworks are proposed in figures 15-16. The position of the shock front is captured at the analytical position and the density in the post-shock region is equal to the analytical value 4. However, there are several differences between both numerical profiles: first, the wall-heating region is larger in the ALE framework, which may be explained by a loss of accuracy near the moving wall due to a less refined mesh. In the Lagrangian framework, excluding the density values near the boundaries, the cell-centered densities match nearly perfectly with the analytical value in the region $0.75 \leq x \leq 0.9$, which is not the case in the ALE frame-

work. In the latter, the range in the density values near the shock seems to be larger. All these difference can be partially explained by the amount of diffusion introduced in the ALE formulation from the upwind numerical fluxes. Results also indicate that the choice of the grid velocities should be made in agreement with the underlying physics, for instance based on the flow velocity.

3.6. Single material Triple point

This problem is a three state single species 2D Riemann problem in a vessel which is inspired from the reference [32]. The computational domain $\Omega = [0; 7] \times [0; 3]$ is given in figure 17. It is split into three subdomains filled with a perfect gas with $\gamma = 1.4$ with different densities and pressures. The triple point is located at $\vec{x} = (1.5, 1.5)^t$. The high pressure high density state in the region $[0; 1.5] \times [0; 3]$ is defined by $(\rho, p) = (1, 1)$, the low pressure low density in the region $[1.5; 7] \times [0; 1.5]$ is $(\rho, p) = (0.125, 0.1)$, and the low pressure high density state in the region $[1.5; 7] \times [1.5; 3]$ is given by $(\rho, p) = (1, 0.1)$. The initial gas is at rest and the final time of the simulation is $t_{\text{final}} = 5$. The mesh is composed of 140×60 square cells.

For a purely Lagrangian method, one expects that the mesh follows the flow with a vortical velocity field, leading to a tangling that ends the simulation very early. The velocity should be then limited to avoid an early crash of the simulation. The result proposed in figure 18 has been obtained by imposing $\forall r \in \mathcal{N}, \vec{w}_r = 0.3 \vec{v}_r$. The mesh distortion is controlled. In figure 19, the mesh deformation in the vortex region is clearly observed.

3.7. Toward an adaptative computational method for the grid velocities

In the previous results, we set the grid velocities to a fraction of the Lagrangian velocities in order to demonstrate the robustness of our scheme. This is clearly not a good strategy in an industrial context, where the computational method of the grid velocities is expected to be automatic. According to us, the most powerful method will be the one for which the grid velocities values adapt to the main characteristics of the flow. Simply, we want the divergence-free part of the velocity to be limited in a vortex, while the irrotational part should be limited in expansion or shocks.

For that purpose, a recent method called L.E.L., for Large Eddy Simulation, has been proposed by Costes *et al.* in reference [1]. The basic idea of the method is to decompose the nodal Lagrangian velocities with the Hodge decomposition [47, 48, 49]:

$$\vec{v}_r = \vec{\nabla} g_r + \vec{\nabla}^\perp f_r, \quad (51)$$

where

$$\nabla^\perp f_r \equiv \left(-\frac{\partial f}{\partial y}, \frac{\partial f}{\partial x} \right), \quad (52)$$

and, for a sake of simplicity, we will assume that $(f, g) \in \mathcal{C}^1(\Omega, \mathbb{R})$. The main feature of the decomposition is to separate the irrotational part of the nodal Lagrangian velocity $\vec{\nabla} g_r$ of the divergence-free part $\vec{\nabla}^\perp f_r$. Existence, uniqueness and the method to compute functions f_r and g_r are discussed in [1].

Once the decomposition is performed, the grid velocity is expressed as a limited form of the Hodge's decomposition of the Lagrangian velocity:

$$\forall r \in \mathcal{N}, \quad \vec{w}_r = \eta_r \vec{\nabla} g_r + \theta_r \vec{\nabla}^\perp f_r, \quad (53)$$

where η_r and θ_r are limiters that may be expressed as:

$$\eta_r = \varphi_{\chi_g}(|\vec{\nabla} g_r|), \quad (54)$$

$$\theta_r = \varphi_{\chi_f}(|\vec{\nabla}^\perp f_r|). \quad (55)$$

Several choices may be made for the limiter functions φ_{χ_g} and φ_{χ_f} . As proposed in [1], we take in this paper:

$$\varphi_{\chi_g}(m) = \max(0, \min(\frac{\chi_g}{m}, 1)), \quad (56)$$

$$\varphi_{\chi_f}(m) = \max(0, \min(\frac{\chi_f}{m}, 1)), \quad (57)$$

where the real numbers χ_g and χ_f denote the limitation thresholds respectively for the irrotational part and the divergence-free part of the flow velocity. They have been chosen constant, implying that the limitation is performed uniformly for each node within the mesh. Note that, if $\chi_g = \chi_f = 0$, then the grid velocities are equal to zero in the whole computational domain, and the simulation is performed in the Eulerian framework. If $\chi_g = |\vec{\nabla} g_r|$ and $\chi_f = |\vec{\nabla}^\perp f_r|$ for every node, then the simulation is Lagrangian and the mesh will follow the flow. By cancelling only one of the two thresholds, either the divergence-free part or the irrotational part of the flow is cut.

In reference [1], the L.E.L. method has been coupled with an indirect ALE hydrocode and shows its ability to adapt the grid velocity to the characteristics of the flow. Our intention is to couple the L.E.L. method to our

numerical scheme to obtain a full automatic direct ALE solver. We propose in figure (20) a preliminary result obtained for the triple point test case by combining our DALEC³ hydrocode with the L.E.L. method. For this result, we set $\chi_g = 0.8$ and $\chi_f = 0.6$. The coefficients have been chosen to limit the early distortion of the mesh by cutting the effect of the vortex. The solution is similar to the one in figure 18, but we observe a pinch in the mesh in the arm of the vortex, which still remains unexplained.

4. Conclusions and Perspectives

In this paper, we have presented in detail a multidimensional Direct ALE scheme for the numerical solution to the compressible Euler equations. The scheme has been developed from a Godunov-like cell-centered Lagrangian scheme, like GLACE or EUCCLHYD, which has been supplemented with upwinded advective fluxes. The upwinded values are chosen from the sign of the edge relative velocity $\vec{\mu} \cdot \vec{\nu}$. The second-order extension is obtained by computing the numerical fluxes from the MUSCL-Hancock method. The scheme is explicit in time. The 3D extension is straightforward.

The robustness of the scheme has been investigated in several test cases, in which the grid velocities have been explicitly defined. Our objective is to obtain a fully adaptive computational method for the grid velocities, and we choose here to couple our scheme with the Large Eddy Limitation method recently proposed by Costes *et al.*. A promising result has been obtained on the single material triple point problem: the method shows its ability to limit the early strong rotation of the mesh induced by the vortex and the numerical solution is close to what was expected.

In forthcoming works, we will present the results obtained by a full coupling with the L.E.L. method, and our efforts will be focused on the extension of the scheme to multi-material frameworks.

Acknowledgement

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Appendix A. Derivation of the semi-discrete formulation of the entropy balance

In this section, a proof for Proposition 2 is given. For the sake of clarity, we will derive the entropy balance for the first order scheme, that is the boundary extrapolated quantities are chosen as the cell-centered quantities (i.e. gradients are zero). We start from the Gibbs relation:

$$Tds = de + pd\tau, \quad (\text{A.1})$$

where e and τ respectively denote the specific internal energy and the specific volume ($\tau = 1/\rho$) of the domain $V(t)$. The semi-discrete time formulation of the relation (A.1) is:

$$\rho_j V_j T_j \frac{ds_j}{dt} = \rho_j V_j \frac{de_j}{dt} + \rho_j V_j p_j \frac{d\tau_j}{dt}.$$

The expression for the first term of the right-hand side is given from relations (31) and (32):

$$\begin{aligned}\rho_j V_j \frac{de_j}{dt} &= \rho_j V_j \frac{dE_j}{dt} - \rho_j V_j \vec{v}_j \cdot \frac{d\vec{v}_j}{dt}, \\ &= \frac{d(\rho_j V_j E_j)}{dt} - \vec{v}_j \cdot \frac{d(\rho_j V_j \vec{v}_j)}{dt} - (E_j - \vec{v}_j^2) \frac{d(\rho_j V_j)}{dt}.\end{aligned}$$

Relations (29), (31) and (32) give:

$$\begin{aligned}\rho_j V_j \frac{de_j}{dt} &\approx - \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} \left[E_k^{\text{up}} - \vec{v}_j \cdot \vec{v}_k^{\text{up}} - E_j + \vec{v}_j^2 \right] \vec{\mu}_k \cdot \vec{\nu}_{j,k} \\ &\quad - \sum_{k \in \mathcal{E}(j)} \left(W_{j,k} - \vec{F}_{j,k} \cdot \vec{v}_j \right).\end{aligned}$$

Furthermore:

$$\begin{aligned}\rho_j V_j p_j \frac{d\tau_j}{dt} &= - \frac{p_j}{\rho_j} \left(\frac{d(\rho_j V_j)}{dt} - \rho_j \frac{dV_j}{dt} \right) \\ &\approx \frac{p_j}{\rho_j} \left(\sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} \vec{\mu}_k \cdot \vec{\nu}_{j,k} + \rho_j \sum_{k \in \mathcal{E}(j)} l_k \vec{w}_k \cdot \vec{\nu}_{j,k} \right).\end{aligned}$$

Finally, we obtain the following relation:

$$\begin{aligned}\rho_j V_j T_j \frac{ds_j}{dt} &\approx - \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} \left[E_k^{\text{up}} - \vec{v}_j \cdot \vec{v}_k^{\text{up}} - E_j + (\vec{v}_j)^2 - \frac{p_j}{\rho_j} \right] \vec{\mu}_k \cdot \vec{\nu}_{j,k} \\ &\quad - \sum_{k \in \mathcal{E}(j)} \left(W_{j,k} - \vec{F}_{j,k} \cdot \vec{v}_j \right) + p_j \sum_{k \in \mathcal{E}(j)} l_k \vec{w}_k \cdot \vec{\nu}_{j,k}.\end{aligned}\tag{A.2}$$

Then, a simple calculation shows that:

$$E_k^{\text{up}} - \vec{v}_j \cdot \vec{v}_k^{\text{up}} - E_j + (\vec{v}_j)^2 = e_k^{\text{up}} - e_j^n + \frac{1}{2} \left(\vec{v}_k^{\text{up}} - \vec{v}_j \right)^2.$$

We modify the first term on the right-hand side of (A.2) as:

$$\begin{aligned}
 & - \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} \left[E_k^{\text{up}} - \vec{v}_j \cdot \vec{v}_k^{\text{up}} - E_j + (\vec{v}_j)^2 - \frac{p_j}{\rho_j} \right] \vec{\mu}_k \cdot \vec{\nu}_{j,k} = \\
 & - \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} \left[e_k^{\text{up}} - e_j + \frac{1}{2} \left(\vec{v}_k^{\text{up}} - \vec{v}_j \right)^2 + p_j \left(\frac{1}{\rho_k^{\text{up}}} - \frac{1}{\rho_j} \right) \right] \vec{\mu}_k \cdot \vec{\nu}_{j,k} \quad (\text{A.3}) \\
 & + p_j \sum_{k \in \mathcal{E}(j)} l_k \vec{\mu}_k \cdot \vec{\nu}_{j,k}.
 \end{aligned}$$

We may notice that the expression

$$e_k^{\text{up}} - e_j + p_j \left(\frac{1}{\rho_k^{\text{up}}} - \frac{1}{\rho_j} \right),$$

stands for a first-order spatial discretization of the Gibbs relation (A.1),

leading to:

$$e_k^{\text{up}} - e_j + p_j \left(\frac{1}{\rho_k^{\text{up}}} - \frac{1}{\rho_j} \right) = T_j (s_k^{\text{up}} - s_j).$$

We gather the last terms of (A.3) and (A.2):

$$p_j \sum_{k \in \mathcal{E}(j)} l_k \vec{\mu}_k \cdot \vec{\nu}_{j,k} + p_j \sum_{k \in \mathcal{E}(j)} l_k \vec{w}_k \cdot \vec{\nu}_{j,k} = p_j \sum_{k \in \mathcal{E}(j)} l_k \vec{v}_k \cdot \vec{\nu}_{j,k},$$

and we need to reformulate this sum at the nodes of the mesh:

$$p_j \sum_{k \in \mathcal{E}(j)} l_k \vec{v}_k \cdot \vec{\nu}_{j,k} = p_j \sum_{r \in \mathcal{N}(j)} l_{j,r} \vec{v}_r \cdot \vec{\nu}_{j,r}. \quad (\text{A.4})$$

The sum $\sum_{k \in \mathcal{E}(j)} \left(W_{j,k} - \vec{F}_{j,k} \cdot \vec{v}_j \right)$ is also reformulated at the nodes:

$$\sum_{k \in \mathcal{E}(j)} \left(W_{j,k} - \vec{F}_{j,k} \cdot \vec{v}_j \right) = \sum_{r \in \mathcal{N}(j)} \left(W_{j,r} - \vec{F}_{j,r} \cdot \vec{v}_j \right),$$

and we obtain the following relation by using the definition of $\overrightarrow{F_{j,k}}$ (15):

$$\begin{aligned}
 \sum_{k \in \mathcal{E}(j)} \left(W_{j,k} - \overrightarrow{F_{j,k}} \cdot \overrightarrow{v_j} \right) &= - \sum_{r \in \mathcal{N}(j)} l_{j,r} p_j \overrightarrow{\nu_{j,r}} \cdot (\overrightarrow{v_r} - \overrightarrow{v_j}) \\
 &\quad + \sum_{r \in \mathcal{N}(j)} \overline{\overline{\sigma_{j,r}}} (\overrightarrow{v_r} - \overrightarrow{v_j}) \cdot (\overrightarrow{v_r} - \overrightarrow{v_j}) \\
 &= - \sum_{r \in \mathcal{N}(j)} l_{j,r} p_j \overrightarrow{\nu_{j,r}} \cdot \overrightarrow{v_r} + \sum_{r \in \mathcal{N}(j)} \overline{\overline{\sigma_{j,r}}} (\overrightarrow{v_r} - \overrightarrow{v_j}) \cdot (\overrightarrow{v_r} - \overrightarrow{v_j}) .
 \end{aligned} \tag{A.5}$$

The last simplification in the first term of the right-hand side comes from the fundamental geometrical property in GLACE and EUCCLHYD (8). By combining finally (A.4) to (A.5), relation (A.2) turns into:

$$\begin{aligned}
 \rho_j V_j T_j \frac{ds_j}{dt} &\approx -T_j \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} (s_k^{\text{up}} - s_j) \overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} - \frac{1}{2} \sum_{k \in \mathcal{E}(j)} l_k \left(\overrightarrow{v_k^{\text{up}}} - \overrightarrow{v_j} \right)^2 \overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} \\
 &\quad + \sum_{r \in \mathcal{N}(j)} \overline{\overline{\sigma_{j,r}}} (\overrightarrow{v_r} - \overrightarrow{v_j}) \cdot (\overrightarrow{v_r} - \overrightarrow{v_j}) .
 \end{aligned} \tag{A.6}$$

Gathering the first terms on the right and left-hand sides gives the semi-discrete formulation of the entropy balance:

$$\begin{aligned}
 \rho_j V_j T_j \frac{ds_j}{dt} &+ T_j \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} (s_k^{\text{up}} - s_j) \overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} \\
 &= T_j \frac{d(\rho_j V_j s_j)}{dt} - T_j s_j \frac{d(\rho_j V_j)}{dt} + T_j \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} (s_k^{\text{up}} - s_j) \overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} \\
 &= T_j \frac{d(\rho_j V_j s_j)}{dt} + T_j \sum_{k \in \mathcal{E}(j)} l_k \rho_k^{\text{up}} s_k^{\text{up}} \overrightarrow{\mu_k} \cdot \overrightarrow{\nu_{j,k}} .
 \end{aligned}$$

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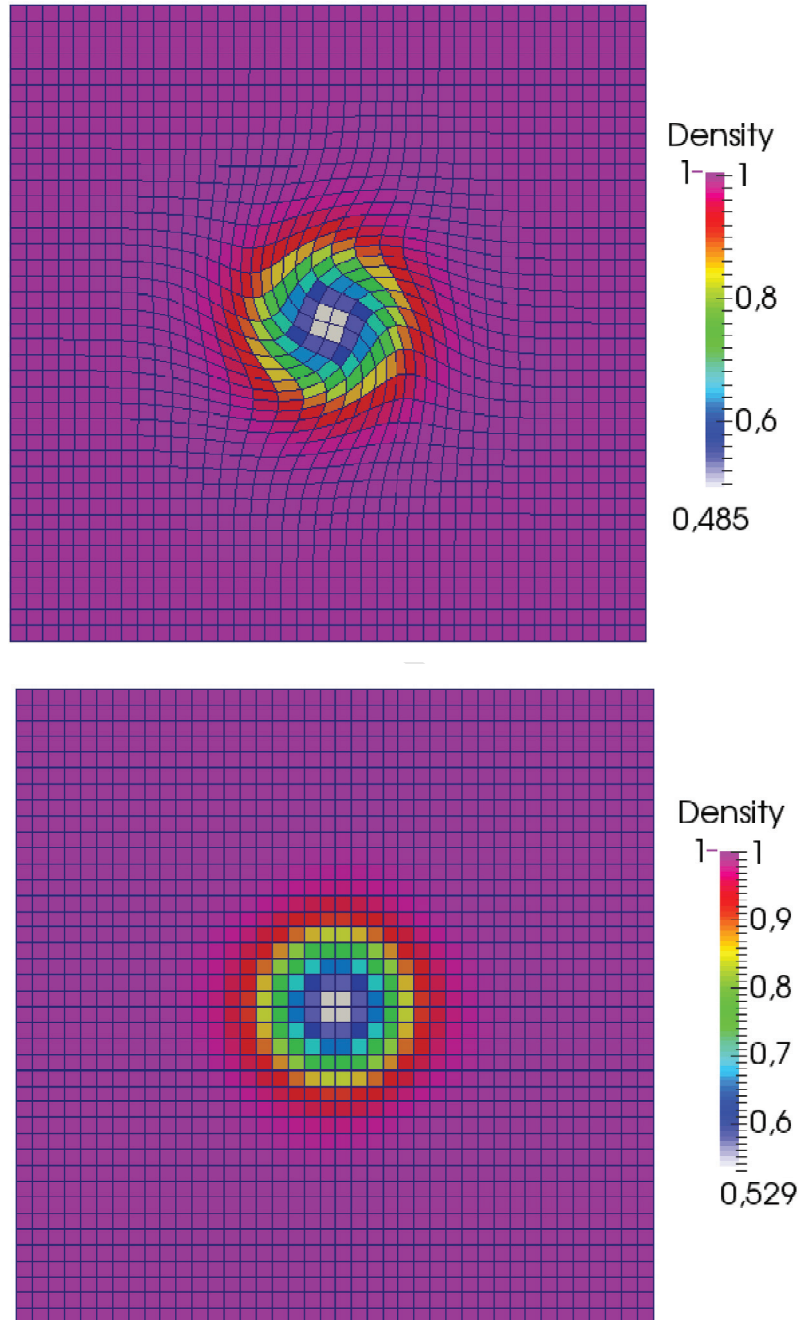


Figure 11: Density map, centered on the vortex eye, computed at time $t = 1$ for the 2D isentropic vortex test case: Lagrangian numerical solution (top) and Eulerian solution (bottom).

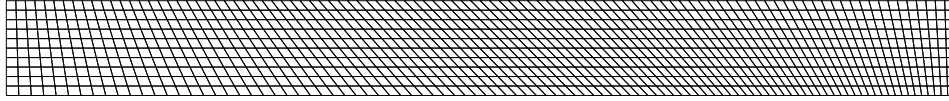


Figure 12: Initial mesh for the Saltzman piston problem

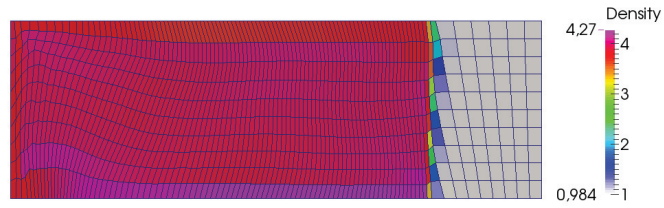


Figure 13: Saltzman problem. Mesh+Density map obtained in the Lagrangian Framework computed at $t = 0.7$

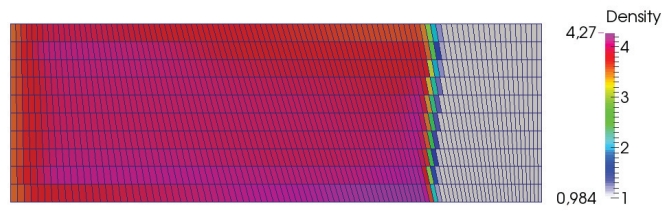


Figure 14: Saltzman problem. Mesh+Density map obtained in the ALE framework computed at $t = 0.7$

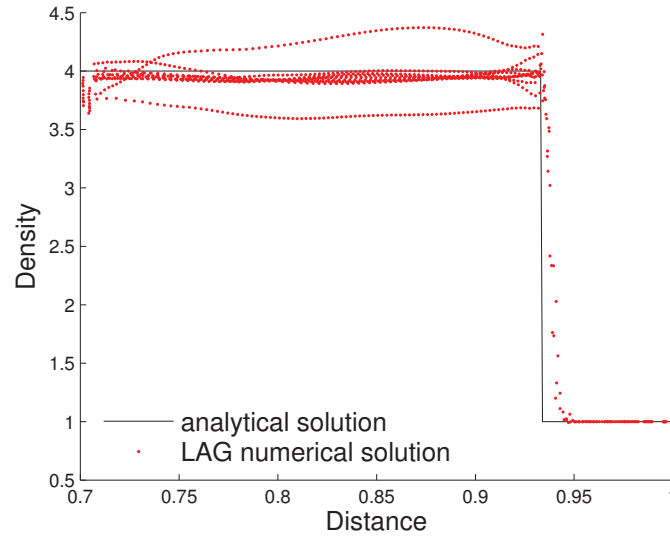


Figure 15: Density in all the cells as function of the cell center x-coordinate versus analytical solution at the computational time $t = 0.7$ - Lagrangian Framework

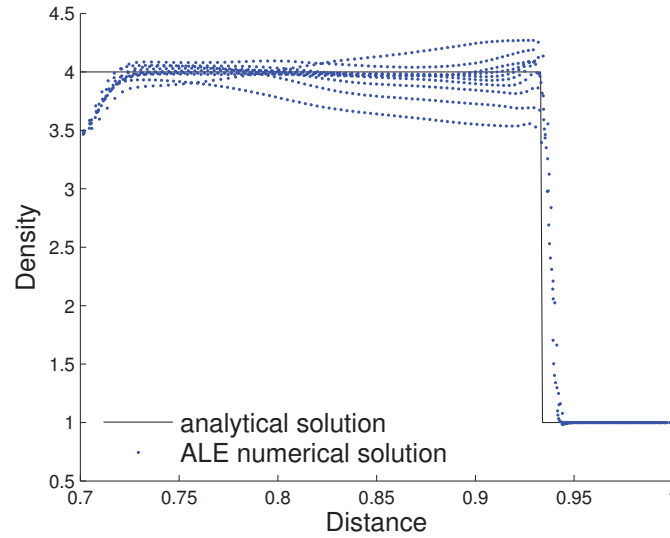


Figure 16: Density in all the cells as function of the cell center x-coordinate versus analytical solution at the computational time $t = 0.7$ - ALE Framework

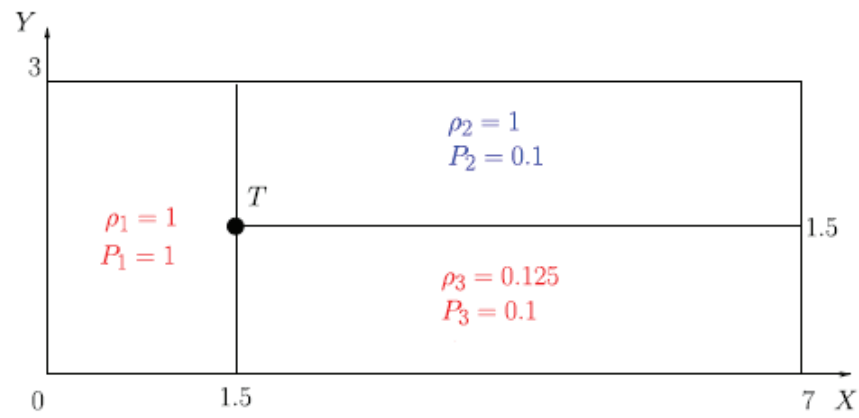


Figure 17: Initialization of the single material triple point problem

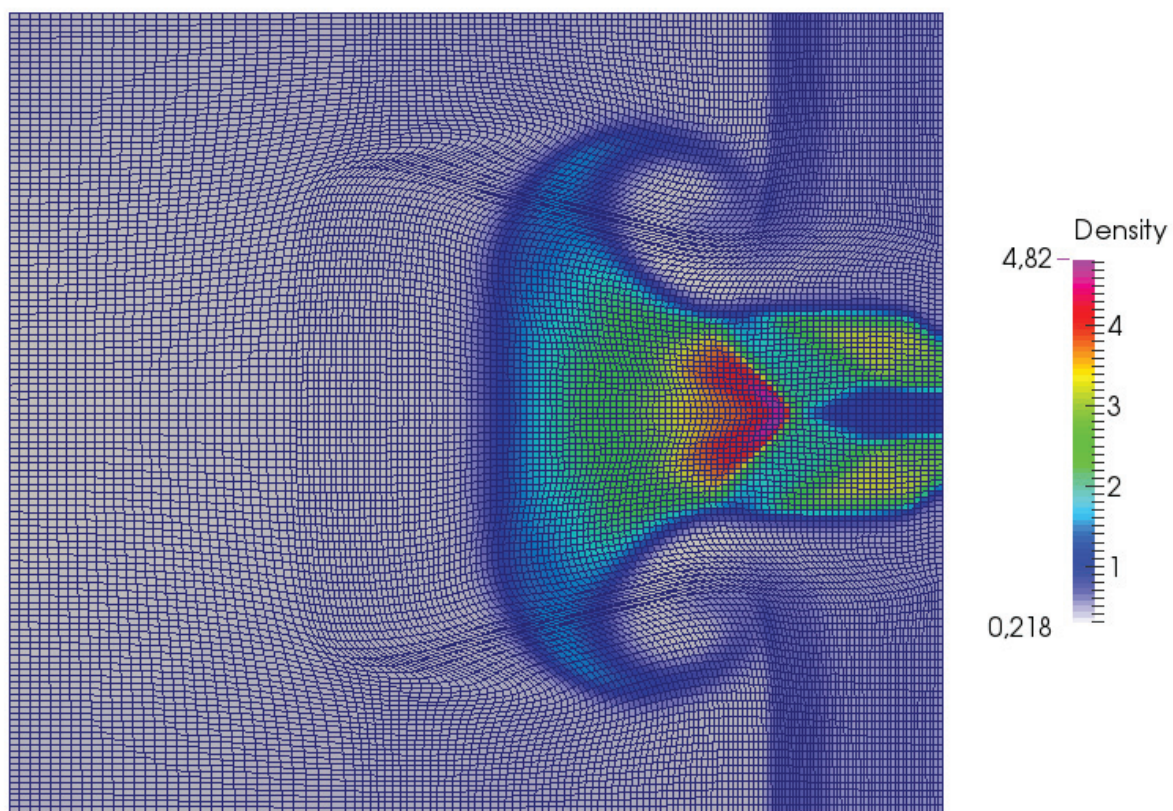


Figure 18: Density map and Mesh computed for the single material triple point test case at the computational time $t_{\text{final}} = 5$. Standard ALE computation with $\vec{w}_r = 0.3 \vec{v}_r$

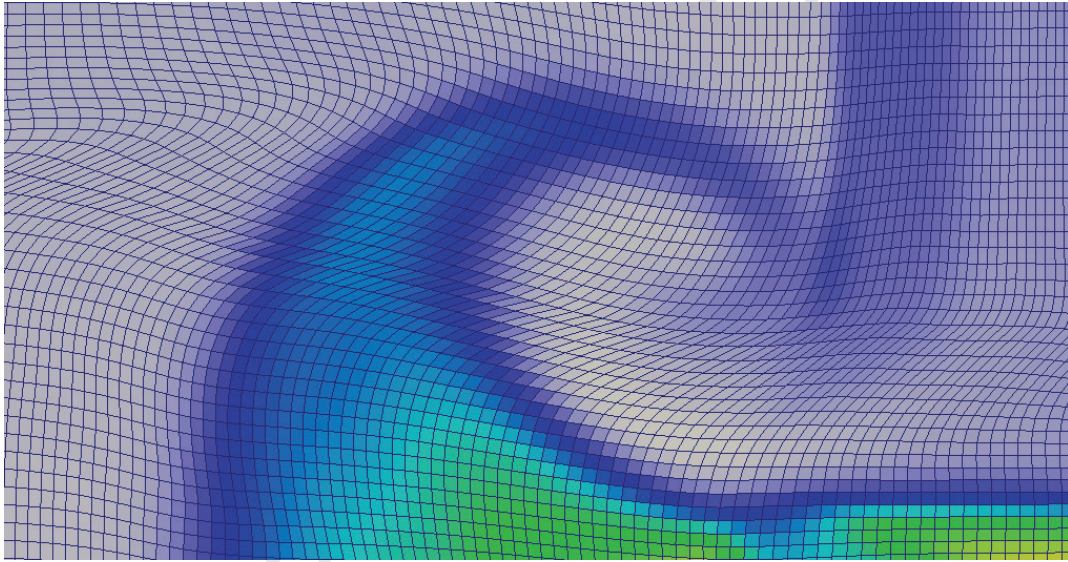


Figure 19: Zoom on the vortex region in the single material triple point test case at the computational time $t_{\text{final}} = 5$.

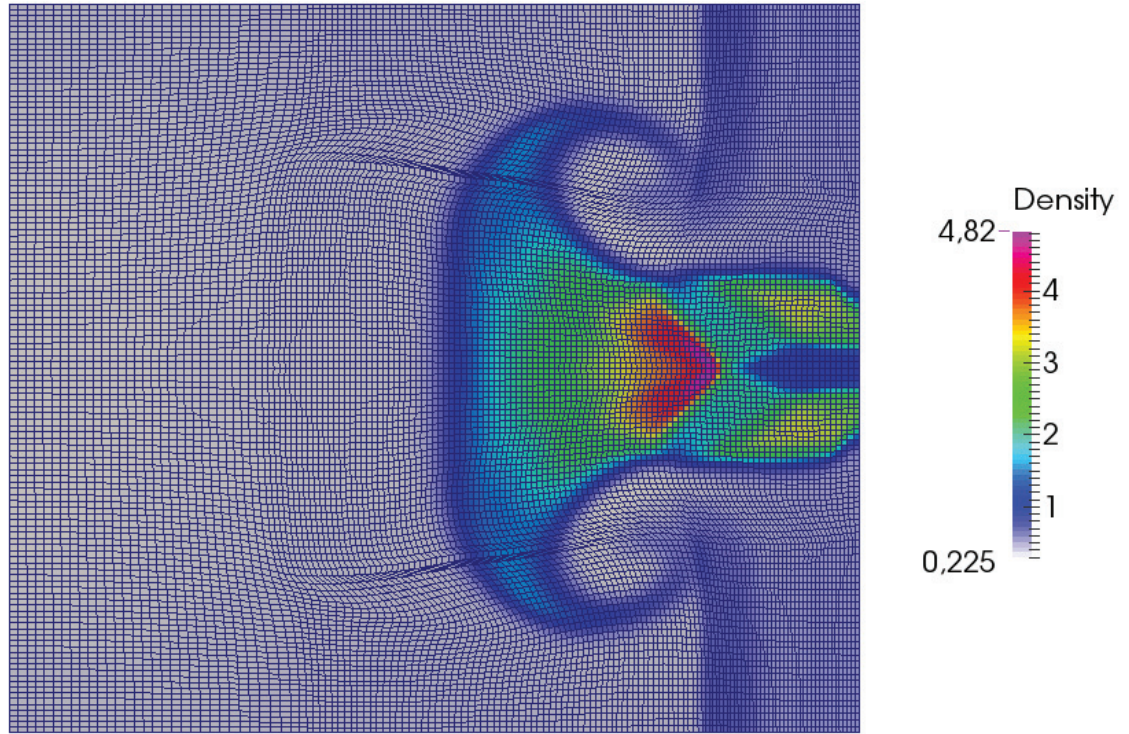


Figure 20: Density map and Mesh computed for the single material triple point test case at the computational time $t_{\text{final}} = 5$. ALE computation with the L.E.L. method.

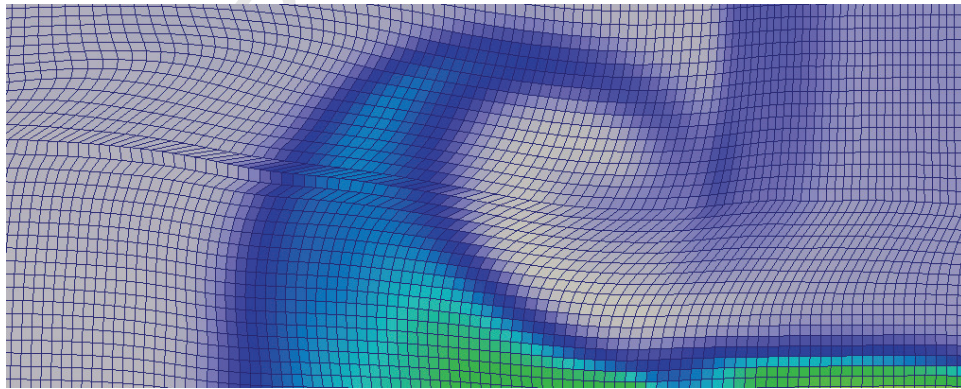


Figure 21: Zoom on the vortex region in the single material triple point test case at the computational time $t_{\text{final}} = 5$. ALE computation with the L.E.L. method.